BINARY SEARCH:

Binary Search is a searching algorithm that finds an element in a sorted array by repeatedly dividing the search range into two halves until the element is found.

ALGORITHM:

1. Sort the array (if it is not already sorted).

2.Find the middle element of the array.

3.Compare the middle element with the search key.

4.If both are equal, the element is found.

5.If the search key is smaller, search in the left half.

6.If the search key is larger, search in the right half.

7.Repeat the process until the element is found or the search range becomes empty.

Time Complexity:

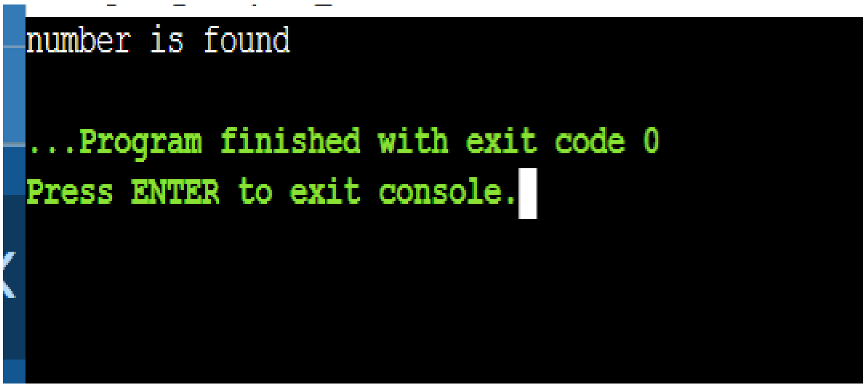
Case	Time Complexity
Best Case	**O(1)**
Average Case	**O(log n)**
Worst Case	**O(log n)**

Space Complexity:

Iterative Binary Search **= O(1)**

Recursive Binary Search **= O(log n)**

OUTPUT:


number is found
...Program finished with exit code 0
Press ENTER to exit console.